

Mohanna Hoveyda

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RESEARCH INTERESTS

Conversational AI, Information Retrieval, Probabilistic logic and constraint solvers, LLM-based reasoning, Reinforcement learning for multi-agent systems.

EDUCATION

PhD, Radboud University, iCIS Nijmegen, The Netherlands
Knowledge-Enriched Conversational AI *February 2023–Present*
Advisors: Arjen P. de Vries, Faegheh Hasibi, Maarten de Rijke
Research Group: Informagus

MSc, University of Tehran, FNST Tehran, Iran
Computational Linguistics *September 2020–January 2023*
Thesis: Network Science-Based Structural Analysis of Human- and Machine-Based Semantic Representations

Bachelor, University of Tehran, ECE & FLL Tehran, Iran
Computer Engineering (Minor) *2018–2020*
French Language and Literature (Major) *2016–2020*

RESEARCH

OrLog: Resolving Complex Queries with LLMs and Probabilistic Logic Programming **In Preparation**

Mohanna Hoveyda, Jelle Piepenbrock, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Developed a retrieval framework that introduces a *probabilistic prior estimator* combined with a *logic programming engine* for constraint-aware inference in face of queries with disjunction, and negation.
- Higher retrieval accuracy and ~90% lower token cost vs. end-to-end LLM reasoning. Results hint at avenues for reasoning with smaller LLMs as alternatives to brute-force scaling.

Adaptive Orchestration of Modular Generative Information Access Systems **SIGIR 2025**

Mohanna Hoveyda, Harrie Oosterhuis, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Designed a modular GenIA architecture for real-time adaptive orchestration of *LLM agents*, retrieval models, and external tools per query to maximize performance while cutting computational overhead.
- Developed a dynamic QA system that adaptively selects optimal answering strategies based on question complexity, using *contextual multi-armed bandits*.

Real-World Conversational Entity Linking Requires More Than Zero-Shots **ACL 2024 Findings**

Mohanna Hoveyda, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Introduced benchmark and evaluation scenarios for a more realistic assessment of zero-shot entity linking models.

Network Science-Based Structural Analysis of Human and Machine-Based Semantic Representations **Master's Thesis**

Mohanna Hoveyda, Paulino Villas-Boas, Mahmood Bijankhan, Mostafa Salehi

- Designed a statistical framework with configuration models and hypothesis testing for size-invariant comparison and clustering of non-isomorphic networks.
- Applied graph-theoretic metrics to compare semantic representations across machine-learned (uni-modal: Word2Vec, BERT; multi-modal: VisWord2Vec), human-curated (WordNet, ConceptNet), and hybrid (ConceptNet Numberbatch) spaces.

PRESENTATIONS

Understanding Diverse Reasoning Procedures in Foundation Models via Mechanistic Interpretability. Poster presentation, Cognitive Computational Neuroscience Conference (CCN'25).

Extended abstract co-authored with Jasmin Kareem, Roxana Petcu, Angela van Sprang, and Ana Lucic.

Designs of Future Information Access Systems Should Be Modular, Adaptive and Dynamic. Poster presentation, CONSEQUENCES at RecSys'25

SKILLS & TOOLS

Python, C, C++
PyTorch, TensorFlow, Scikit-learn, Hugging Face
ElasticSearch, Lucene, Faiss
HPC, vLLM
Z3 SMT solver, ProbLog
NetworkX, Matplotlib, Seaborn

LANGUAGES

Persian (Native), *English* (Fluent), *French* (Intermediate), *Dutch* (Basic)

TEACHING & SUPERVISION

Teaching Assistant Information Retrieval (Radboud University), Machine Learning for NLP (University of Tehran)

Supervision *Tsvetomira Krikoryan* (MSc internship, Wintertuin Company): *Training a GPT-2 model on Papiamentu for creative task assistance*

AWARDS & SCHOLARSHIPS

Ranked 6th in Computational Linguistics National Master's Entrance Exam, Iran National Organization for Educational Testing (2020)

Scholarship for Computational Linguistics Master's Program, University of Tehran, Iran (2020)

Scholarship for Bachelor of French Literature, University of Tehran, Iran (2016)

WORKSHOPS & SCHOOLS

European Summer School on Information Retrieval (ESSIR), University of Amsterdam, Netherlands (2024); Technische Universität Wien, Austria (2023)

High-Performance Computing (HPC) and Deep Learning Workshop, HPC Laboratory, IPM, Tehran, Iran (2022)

Summer School of Intelligent Learning, IPM & Sharif University of Technology, Tehran, Iran (2019)

Autumn School of Mind and Brain, Cognitive Sciences and Technologies Council & University of Tehran, Tehran, Iran (2018)

REFERENCES

Arjen P. de Vries a.devries@cs.ru.nl
Professor of Information Retrieval at Radboud University and Research Director of iCIS

Maarten de Rijke m.derijke@uva.nl
Professor of Artificial Intelligence and Information Retrieval at UvA